

Quotients in distribution and the leading-order posterior expectation (Programme S, tranche A4-L)

Claude, with Daniel Murfet

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Abstract

Units 286–288 close Programme S with the leading term of `cor:empirical_expectation` after finite chart assembly. The generic statement is probabilistic: if $(Y_\ell, W_\ell) \Rightarrow (A, B)$ jointly and $P(B = 0) = 0$ then $Y_\ell/W_\ell \Rightarrow A/B$ (Lean’s totalised division; nothing about $W_\ell \neq 0$ at finite ℓ). Mathlib has no continuous mapping theorem for maps continuous only almost surely, so the proof uses the truncated division $T_\delta(a, b) = ab/\max(b^2, \delta^2)$, globally continuous and equal to a/b on $|b| \geq \delta$: for a bounded Lipschitz test function of oscillation C the integrals of $f(Y/W)$ and $f(T_\delta(Y, W))$ differ by at most $C P(|W| \leq \delta)$, portmanteau bounds $\limsup P(|W_\ell| \leq \delta)$ by $P(|B| \leq \delta)$, and $P(|B| \leq 1/(m+1)) \downarrow P(B = 0) = 0$. The assembled corollary takes the pair (denominator, numerator) of joint chart data, a.e. chart decompositions with residuals negligible at the leading scale $a_\ell = N_\ell^{-\mu_0}(\log N_\ell)^{j_0}$, aggregate predecessor sums vanishing a.s. at every source index, and $P(C_{\mu_0, j_0}^1(X) = 0) = 0$; the normalised pair converges jointly by vector Slutsky and the quotient theorem finishes (Headline XXXVIII). Review v31: pass. Three units against a cap of twelve. No `sorry`, no additional `axiom`; 3415 jobs; pin `a1a7d1a`.

1 Why only the leading term

Astra #35 refused the all-orders quotient expansion: `lemma:division` has no statement in the source, and dividing by $1 + c/\log n$ produces an infinite block of inverse logarithmic powers at a fixed power of n , which the finite predecessor ordering of the Taylor tree cannot host. The leading term, however, is exactly a joint-convergence-plus-division statement, and the assembled Taylor tree already provides the joint convergence.

2 The generic quotient theorem (u286)

The weak convergence of laws is characterised by integrals of bounded Lipschitz test functions (`tendsto_iff_forall_lipschitz_integral_tendsto`, hence the countably generated filter). For such an f with oscillation bound C : (i) $|f(Y/W) - f(T_\delta(Y, W))| \leq C \mathbf{1}_{\{|W| \leq \delta\}}$ pointwise, so the integral difference is at most $C P(|W| \leq \delta)$ (indicator integral on a probability space; integrability from the oscillation bound, $\|f(x)\| \leq \|f(0)\| + C$); (ii) $W_\ell \Rightarrow B$ by projecting the pair, and `ProbabilityMeasure.limsup_measure_closed_le_of_tendsto` on the closed set $\{|b| \leq \delta\}$ gives eventually $P(|W_\ell| \leq \delta) < \eta$ once $P(|B| \leq \delta) < \eta$; (iii) the sets $\{|B| \leq 1/(m+1)\}$ decrease to $\{B = 0\}$, so δ exists (`tendsto_measure_iInter_atTop`); (iv) continuous mapping at fixed δ ; (v) $\varepsilon/3$ with $\eta = \varepsilon/(6(C+1))$. The scaled form removes a common deterministic normalisation $a_\ell \neq 0$ (the cancellation $(U/a)/(V/a) = U/V$ fails at $a = 0$ under totalised division, so $a_\ell \neq 0$ is required at every index).

3 The assembled leading quotient (u287–288)

`PairData` is a `def` synonym of `JointData × JointData` with the sup norm and the Borel σ -algebra (denominator first, numerator second). The type does not enforce a common phase or $\eta_\phi = \phi \cdot \eta_1$; the paper’s model is a specialisation. With the a.e. decompositions and vanishing predecessor sums, $Z_\ell^0/a_\ell - C(X_\ell) = (R_\ell - C(X_\ell)) + E_\ell/a_\ell$ a.e. for both families; the first term is uniformly small on joint balls (the sup norm of the pair dominates both components), the second is the normalised residual; both pairs tend to zero in probability (two small lemmas: sums and pairs of in-probability-zero sequences), and vector Slutsky gives $(Z^0[\phi]/a, Z^0[1]/a) \Rightarrow (C^\phi(Y), C^1(X))$. The reviewer’s points: remainders minus coefficients *at the same source datum* tend to zero (the limit lives on another probability space); aggregate predecessor sums vanish, individual coefficients need not; the numerator coefficient may vanish; $P(C^1 = 0) = 0$ is not positivity; no convergence of expectations follows.

4 Lessons

- **Truncate the map, not the variables.** A globally continuous surrogate agreeing with the discontinuous map off a small-probability set replaces the a.s.-continuous mapping theorem; the test function’s boundedness pays for the bad set.
- **rw versus refine on weak-convergence characterisations.** Rewriting the `Tendsto` goal with the characterisation fails on instance mismatch; `refine (lemma (Ω := )).2 ?_` unifies.
- **Order of the pair.** Store data as (denominator, numerator) and produce coefficients as (numerator, denominator) for division; the reviewer checked this explicitly — document it.
- **Product σ -algebras.** Products of nonseparable function spaces need a declared Borel instance on a synonym; never let `Pi.borelSpace/Prod.borelSpace` be assumed.